Progressive Contrastive Learning With Confidence-Weighted Sample Contribution for Cross-Scene Hyperspectral Image Classification

Haonan Chang, Lina Yang, *Member, IEEE*, Guanyu Qin, Xinzhang Wu, *Senior Member, IEEE*, Jun Pang, and Jing Meng

***Abstract*—Cross-scene hyperspectral image classification (HSIC) under unsupervised domain adaptation (UDA) remains challenging due to substantial distribution discrepancies caused by variations in acquisition conditions. In this setting, labeled samples are only available in the source domain, while the target domain lacks supervision. Despite recent progress, existing hyperspectral image domain adaptation (HSIDA) methods are still vulnerable to severe domain shifts, particularly when unreliable pseudo-labels introduce erroneous supervision. Moreover, unconstrained cross-domain feature interaction may further amplify distribution discrepancies and degrade representation quality. To address these issues, we propose a progressive confidence-weighted contrastive learning network (PC2L-Net) for HSIDA. At its core, a progressive confidence-weighted contrastive learning (PCCL) strategy is developed to regulate target-domain sample contributions based on pseudo-label reliability, mitigating error propagation during contrastive learning. In addition, an LMMD gradient-guided reweighting (LGGR) module performs feature dimension-level discrepancy-aware alignment by adaptively reweighting spectral bands. Furthermore, an interdomain sparse cross-attention (ISCA) mechanism enables controlled cross-domain interaction while suppressing discrepancy amplification. Extensive experiments on three hyperspectral benchmark datasets demonstrate that the proposed method consistently outperforms representative HSIDA methods, yielding more discriminative and robust cross-domain representations.**

***Index Terms*—Contrastive learning, cross-scene, domain adaptation (DA), hyperspectral image classification (HSIC), maximum mean discrepancy (MMD).**

I. Introduction

HYPERSPECTRAL images (HSIs) have become indispensable in remote sensing applications such as land-cover mapping, environmental monitoring, and precision agriculture due to their ability to capture hundreds of contiguous spectral bands [1]. This rich spectral information enables fine-grained material discrimination, facilitating accurate identification of diverse surface types and substances [2]. However, despite these advantages, HSI-based classification still faces a major obstacle in practical deployment—domain shift. Domain shift refers to the degradation in model performance when models trained on one scene are applied to another, where the data distribution of the target domain differs from that of the source domain [3]. Such discrepancies commonly arise from variations in illumination conditions, atmospheric effects, sensor characteristics, and geographic environments [4], leading to significant performance degradation under real-world, dynamic scenarios.

To mitigate this problem, domain adaptation (DA) has emerged as an effective strategy for enhancing the generalization ability of HSI classification (HSIC) models [5], [6], [7], [8]. DA techniques aim to bridge the gap between labeled source-domain data and unlabeled target-domain data by aligning their underlying feature distributions and transferring discriminative knowledge across domains [9]. With the rapid advancement of deep learning, numerous DA methods have been proposed to improve cross-domain robustness by learning domain-invariant features [10], [11]. Despite notable progress, these methods often struggle when the target-domain data are scarce, noisy, or highly heterogeneous—conditions frequently encountered in real-world remote sensing. Such challenges underscore the necessity for more adaptive and discriminative representation learning mechanisms capable of achieving reliable DA under complex and dynamic conditions.

Traditional feature extractors, such as convolutional neural networks (CNNs), have long been central to HSIC due to their strong capability in learning local spectral–spatial patterns [12], [13]. However, the inherent locality of convolution operations leads to restricted receptive fields, which limits the modeling of long-range dependencies and hinders the full exploitation of the high-dimensional spectral characteristics of HSIs. To alleviate these limitations, several studies have explored contextual modeling strategies and deeper spectral–spatial architectures, such as [14] and [15]. Nevertheless, as hyperspectral datasets grow in both spatial extent and spectral dimensionality, the computational cost of CNN-based models increases rapidly, posing challenges to efficiency and scalability [16].

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To overcome the inherent locality of convolution-based models, attention mechanisms have been introduced to enhance feature extraction in HSIC [17]. By adaptively emphasizing informative spectral bands and spatial regions, attention-based architectures are capable of capturing long-range dependencies and subtle interband relationships, thereby substantially improving the discriminative capacity of hyperspectral representations [18]. Nevertheless, most existing attention-based models are primarily developed for single-domain settings, where the objective is to enhance intradomain feature representations. When such mechanisms are directly transferred to DA tasks, the learned attention weights may become biased toward domain-specific patterns, as cross-domain distribution discrepancies are not explicitly regulated. To alleviate this issue, recent studies have incorporated attention modules into cross-domain learning frameworks to model interactions between source and target features [19], [20]. Although these approaches attempt to capture interdomain relationships, unrestricted feature interaction may amplify domain discrepancies and introduce unstable responses during alignment. These limitations indicate the necessity of designing sparsity-constrained cross-domain attention mechanisms that selectively emphasize informative correspondences while suppressing discrepancy amplification. Such controlled interaction is essential for achieving reliable feature exchange and learning domain-invariant yet discriminative spectral–spatial representations.

In recent years, contrastive learning has attracted increasing attention in DA research, as it provides instance-level discriminative supervision that complements distribution-level alignment strategies [21], [22]. By organizing the latent space through comparisons between positive and negative sample pairs, contrastive learning encourages compact intraclass clustering and clear interclass separation, thereby enhancing cross-domain consistency [23], [24]. However, when applied to DA, contrastive learning faces inherent challenges due to the absence of reliable supervision in the target domain [25], [26]. To compensate for this limitation, pseudo-labels are commonly introduced to guide representation learning across domains [27], [28]. Nevertheless, the confidence of pseudo-labels can vary substantially under severe domain shifts, and low-confidence pseudo-labels may introduce erroneous supervision signals that misguide optimization and degrade classification performance [29]. Consequently, it is essential to develop mechanisms that selectively emphasize high-confidence pseudo-labels while suppressing the adverse influence of unreliable ones, ensuring that contrastive learning remains stable and effective for cross-domain representation learning.

Beyond representation structuring and contrastive objectives, distribution alignment remains a fundamental component of DA. Among various approaches, statistical distance–based criteria, such as maximum mean discrepancy (MMD) and its variants, including local MMD (LMMD), have been widely adopted due to their effectiveness in characterizing domain-level distribution shifts [30]. By mapping source and target features into a reproducing kernel Hilbert space (RKHS) and minimizing the discrepancy between their empirical distributions, these methods provide a principled mechanism for global or class-aware alignment. Despite their success, existing MMD-based approaches typically treat feature dimensions uniformly when minimizing cross-domain discrepancy. Given the reliable discrepancy estimation offered by MMD-based criteria, exploring more refined alignment strategies that allow differentiated alignment strength across feature dimensions, guided by discrepancy-related cues, may represent a promising direction for further enhancing DA performance. This perspective is particularly relevant to HSIC, where individual spectral bands exhibit distinct physical properties and statistical behaviors, resulting in heterogeneous cross-domain discrepancies across spectral dimensions. Consequently, incorporating feature-level granularity into domain discrepancy minimization is essential for enabling fine-grained and robust adaptation, allowing the model to selectively emphasize feature dimensions that contribute more significantly to domain shift.

To address the aforementioned challenges, we propose a novel framework termed progressive confidence-weighted contrastive learning network (PC2L-Net). The network integrates three complementary modules to achieve robust cross-domain feature alignment. First, a dual-branch feature extractor (DBFE) independently models spectral and spatial characteristics from both domains and concatenates them to form rich spectral–spatial representations. To enable effective interdomain interaction while suppressing redundant responses, an interdomain sparse cross-attention (ISCA) mechanism is introduced. ISCA selectively emphasizes informative spatial and spectral dependencies, facilitating the learning of more reliable and domain-invariant features. Then, the LMMD gradient-guided reweighting (LGGR) module adaptively adjusts the importance of each spectral dimension according to its contribution to the LMMD loss, enabling the model to focus on feature dimensions with stronger domain discrepancies for fine-grained distribution alignment. Finally, a progressive confidence-weighted contrastive learning (PCCL) module is used to enhance representation consistency. By generating augmented intraclass samples and introducing a confidence-weighted contrastive loss (ConfCL), the model effectively leverages high-confidence pseudo-labels while mitigating the influence of uncertain ones. This design ensures stable and reliable optimization, leading to more discriminative and transferable cross-domain representations.

In summary, the contributions of the study are as follows.

1) An ISCA mechanism is introduced to enhance spatial spectral information exchange between domains. By introducing sparsity into the cross-attention process, ISCA enables the model to emphasize informative interdomain dependencies and suppress redundant responses.
2) An LGGR module is developed to perform fine-grained feature-level alignment. By leveraging LMMD gradients with respect to each spectral dimension, LGGR adaptively reweights features to focus on dimensions with greater interdomain discrepancies.
3) A PCCL strategy is proposed, together with a new ConfCL. This loss adaptively balances the influence of samples based on their confidence, effectively pulling

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same-class samples across domains closer and pushing different-class samples apart.

Extensive experiments on multiple benchmark HSI datasets demonstrate that the proposed PC2L-Net significantly outperforms competitive methods, validating its effectiveness and generalization ability for HSI unsupervised DA (UDA).

II. Related Works

A. Domain Adaption for HSIC

In cross-scene HSIC, DA is essential for mitigating the severe distribution discrepancy between a labeled source domain and an unlabeled target domain. Recently, attention-based architectures have attracted substantial interest in HSIC owing to their ability to capture long-range spectral–spatial dependencies and selectively emphasize informative features. For example, Liu et al. [31] proposed a class-wise adversarial adaptation network that incorporates probability prediction-based MMD to enhance feature alignment across domains. Zhao et al. [32] proposed a dual-branch spectral–spatial attention network that integrates global spectral modeling with spatial attention to improve feature discrimination, while Xu et al. [33] introduced a multiscale cross-level attention framework that aggregates hierarchical contextual cues to enhance spectral–spatial representation learning. While these methods have demonstrated the effectiveness of attention mechanisms in learning informative spectral–spatial representations, most of them primarily focus on intradomain feature modeling. In cross-scene adaptation, however, effective knowledge transfer also relies on sufficient interaction between source and target representations to exploit complementary domain-specific information. Motivated by this observation, our method introduces a sparse cross-domain attention mechanism that explicitly facilitates cross-domain feature interaction while suppressing domain-induced interference, thereby yielding more reliable and transferable representations.

Recent adversarial DA methods have further advanced cross-scene HSIC. For example, Huang et al. [34] proposed a spatial–spectral weighted adversarial adaptation framework to improve feature alignment and target-domain classification, while Huang et al. [35] introduced calibrated prototype learning and dynamic instance convolution to enhance feature discriminability across domains. Although these methods effectively mitigate domain discrepancy through adversarial optimization, they primarily focus on distribution alignment. In contrast, our work further investigates representation discrimination from a contrastive learning perspective, aiming to improve category compactness and separability in cross-scene scenarios.

On the other hand, MMD and its variants remain widely employed in hyperspectral DA due to their generality and compatibility with deep neural networks. Recent extensions typically focus on modifying kernel formulations or incorporating class-wise constraints to refine alignment. For instance, Zhu et al. [36] introduced the local MMD (LMMD), which emphasizes fine-grained conditional alignment through class-level reweighting, and Long et al. [37] proposed joint MMD (JMMD) to align multilayer joint distributions and better preserve transferable semantics.

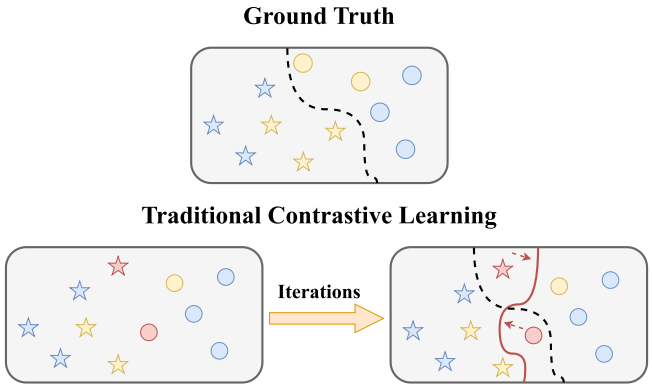


Fig. 1. Limitations of traditional contrastive learning. Blue and yellow denote source- and target-domain samples, respectively, while red indicates incorrectly pseudo-labeled samples.

Despite the substantial progress achieved by MMD-based and adversarial adaptation methods, the role of spectral-dimension heterogeneity in distribution alignment remains relatively less explored. In hyperspectral imagery, different spectral bands often contribute unequally to cross-domain discrepancy due to varying sensitivity to environmental conditions, sensor characteristics, and material properties. Considering such spectral heterogeneity, Li et al. [38] propose the multiteacher multiobjective meta-learning network. This paradigm mines transferable meta-knowledge for band evaluation to select discriminative spectral bands and tackle unseen target domains with limited labeled samples. Motivated by this observation, our method incorporates dimension-wise sensitivity into the alignment objective, enabling more targeted and effective distribution alignment in complex cross-scene scenarios.

B. Contrastive Learning for HSI Domain Adaption

Contrastive learning has been widely adopted in HSIC for its effectiveness in learning discriminative representations by enforcing intraclass compactness and interclass separability. By explicitly modeling similarity relationships in the feature space, contrastive objectives have proven particularly suitable for handling the high spectral variability and complex class distributions inherent in hyperspectral data. Owing to these advantages, several recent studies have extended contrastive learning to HSI UDA, reporting notable improvements in cross-domain feature alignment and classification robustness [39], [40], [41], [42]. For example, Li et al. [43] proposed a guided group contrastive learning framework that integrates supervised signals with unsupervised information to achieve intraclass compactness and interclass separation under limited annotations, while Cai et al. [44] introduced supervised contrastive learning into cross-scene hyperspectral classification by leveraging pseudo-labels to enlarge interclass margins and improve cross-domain discriminability.

However, directly transferring contrastive learning paradigms from HSIC to hyperspectral image domain adaptation (HSIDA) introduces nontrivial challenges. Unlike HSIC, hyperspectral DA typically lacks ground-truth

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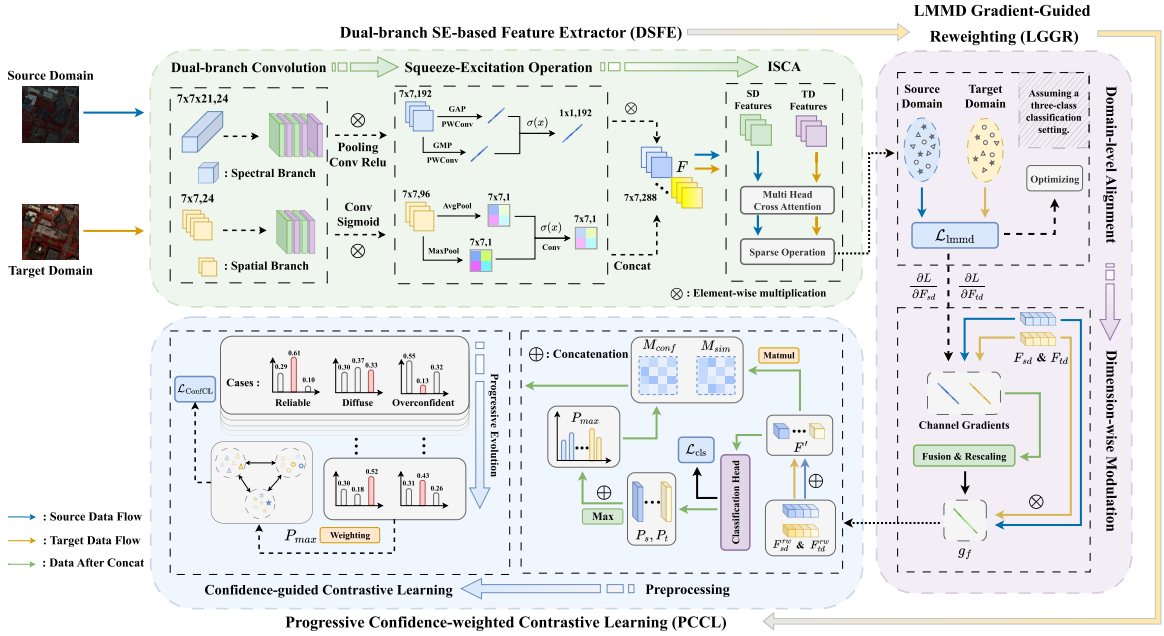


Fig. 2. Overall architecture of the proposed PC2L-Net. Source and target samples are processed by the DSFE, where ISCA is embedded to facilitate cross-domain interaction. The resulting features are refined by LGGR for discrepancy-aware alignment at the feature dimension level, followed by PCCL for PCCL.

annotations in the target domain, compelling existing methods to rely on pseudo-labels to construct positive and negative sample pairs. Consequently, the performance of contrastive learning becomes highly sensitive to pseudo-label quality. Under severe domain shifts, inaccurate pseudo-labels may introduce erroneous supervision signals, which can misguide contrastive optimization and degrade feature discriminability. This limitation is illustrated in Fig. 1. Target-domain samples located near the decision boundary are particularly susceptible to pseudo-label ambiguity. As shown, the target-domain sample marked in red has already been assigned an incorrect pseudo-label during early training stages. When treated equally with other target-domain samples under conventional contrastive learning, such erroneous supervision tends to be progressively reinforced through iterative updates. As a result, samples from the same class may be incorrectly separated, while samples from different classes may be erroneously clustered together, leading to distorted feature structures, as shown in the lower right subfigure of Fig. 1.

These observations indicate that effective contrastive learning for HSIDA should not only leverage pseudo-labels to enhance feature discrimination, but also explicitly consider their reliability. Motivated by this insight, our method adopts a confidence-aware strategy to regulate the influence of pseudo-labeled target samples during contrastive optimization, thereby enabling a more stable and progressive alignment process under large domain discrepancies.

III. Proposed Method

The overall architecture of the proposed PC2L-Net is illustrated in Fig. 2. The framework aims to enhance cross-domain robustness in HSIC by progressively improving feature interaction, alignment, and representation consistency. As shown, the network first employs a dual-branch SE-based feature extractor (DSFE) that independently extracts spectral and spatial representations from both source and target domains, which are then concatenated to form comprehensive joint features. These features are refined through the ISCA module, enabling effective cross-domain information exchange while suppressing redundant responses. The LGGR module further emphasizes feature dimensions with larger interdomain discrepancies to enhance distribution alignment. Finally, the confidence-aware contrastive learning (CACL) module imposes a confidence-weighted contrastive constraint to achieve stable and discriminative representation learning across domains.

A. Dual-Branch SE-Based Feature Extractor

As shown in the green dashed module in Fig. 2, the DSFE, motivated by [20], is adopted as the backbone of PC2L-Net to jointly capture the spectral and spatial characteristics of HSIs. Given the input samples from the source and target domains, denoted as $\mathbf{X}_s$ and $\mathbf{X}_t$, DSFE produces fused spectral–spatial representations as

$$\mathbf{F} = \mathrm{DSFE}(\mathbf{X}) \tag{1}$$

where $\mathbf{F} \in \mathbb{R}^{H\times W\times C'}$ denotes the extracted feature maps.

1) Spectral and Spatial Feature Extraction: DSFE consists of two parallel branches that explicitly decouple spectral and spatial modeling. The spectral feature extraction branch is designed to capture interband dependencies among contiguous spectral channels. By employing 3-D convolutional operations, this branch progressively aggregates spectral correlations while preserving spatial resolution, enabling effective modeling of high-dimensional spectral information that is critical for material discrimination in HSIs.

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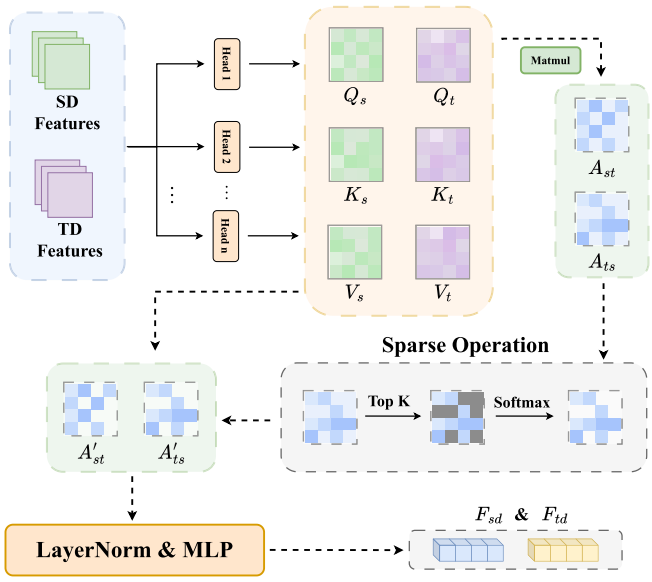


Fig. 3. Illustration of ISCA. Gray color in sparse operation represents the value of $-\infty$, while white indicates the value of 0.

In parallel, the spatial feature extraction branch focuses on learning local spatial structures and contextual relationships. This branch utilizes convolutional operations in the spatial domain to enhance sensitivity to spatial patterns, object boundaries, and neighborhood consistency. Through this design, DSFE is able to simultaneously encode complementary spectral and spatial cues, which together form a robust foundation for subsequent cross-domain representation learning.

2) Spectral and Spatial SE: To further improve feature discriminability, lightweight squeeze-and-excitation (SE) mechanisms are incorporated into the spectral and spatial branches. The spectral SE module adaptively recalibrates channel-wise responses by exploiting global spectral statistics, allowing informative spectral bands to be emphasized while suppressing redundant or less discriminative ones. Meanwhile, the spatial SE module refines spatial feature responses by highlighting salient spatial regions and attenuating background noise.

These SE operations enhance the quality of spectral and spatial representations in a parameter-efficient manner, without introducing excessive computational overhead. Finally, the refined features from both branches are concatenated along the channel dimension to form a unified spectral–spatial representation, which serves as the input for subsequent interdomain alignment and contrastive learning modules.

B. Interdomain Sparse Cross-Attention

To facilitate efficient feature interaction and alignment between the source and target domains, the proposed ISCA module establishes bidirectional attention links that selectively emphasize the most relevant interdomain correspondences while suppressing redundant dependencies. As illustrated in Fig. 3, given the source-domain and target-domain feature embeddings $\mathbf{F}_s$ and $\mathbf{F}_t$, ISCA employs a multihead cross-attention structure to enhance interdomain feature representations. Each feature is first normalized using a layer normalization (LN) layer, after which linear projections are applied to generate query, key, and value matrices for both domains

$$\mathbf{Q} = \mathbf{F}\mathbf{W}_{\mathrm{Q}}, \quad \mathbf{K} = \mathbf{F}\mathbf{W}_{\mathrm{K}}, \quad \mathbf{V} = \mathbf{F}\mathbf{W}_{\mathrm{V}} \tag{2}$$

where $\mathbf{W}_{\mathrm{Q}}$, $\mathbf{W}_{\mathrm{K}}$, and $\mathbf{W}_{\mathrm{V}}$ are learnable projection matrices. The projected query, key, and value features are evenly divided into h attention heads to enable parallel subspace interactions. Accordingly, $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are represented as tensors of dimension $\mathbb{R}^{B\times h\times c}$, where B denotes the batch size, h is the number of attention heads, and $c = d/h$ is the channel dimension of each head. The cross-domain attention maps are then computed through element-wise interactions between the corresponding query–key pairs

$$\mathbf{A}_{\mathrm{st}} = \frac{\mathbf{Q}_{\mathrm{s}} \odot \mathbf{K}_{\mathrm{t}}}{\sqrt{c}}; \quad \mathbf{A}_{\mathrm{ts}} = \frac{\mathbf{Q}_{\mathrm{t}} \odot \mathbf{K}_{\mathrm{s}}}{\sqrt{c}} \tag{3}$$

where $\odot$ denotes element-wise multiplication performed independently within each attention head.

To improve computational efficiency and emphasize the most informative cross-domain dependencies, ISCA applies a sparse selection operation that retains only the top-k elements of each attention map based on their magnitude, setting the remaining entries to $-\infty$. The value of k is determined by a predefined sparsity ratio $\rho \in (0, 1]$, such that $k = \lfloor \rho \times N \rfloor$, where N = c denotes the number of channel-wise attention responses within each attention head. The Top-K operation is applied independently along the channel dimension of each attention head

$$\begin{aligned} \mathbf{A}'_{\mathrm{st}} &= \mathrm{Softmax}\left(\mathrm{TopK}\left(\mathbf{A}_{\mathrm{st}}\right)\right) \\ \mathbf{A}'_{\mathrm{ts}} &= \mathrm{Softmax}\left(\mathrm{TopK}\left(\mathbf{A}_{\mathrm{ts}}\right)\right). \end{aligned} \tag{4}$$

In cross-domain attention, especially when the source and target domains are not well-aligned, spurious correlations and noise may arise due to the inherent differences between feature distributions. If these noisy interactions are left unchecked, they could dominate the aggregation process, leading to inaccurate feature alignment. To address this, the sparsification operation effectively mitigates the amplification of noise by filtering out low-confidence interactions, ensuring that only the most reliable and semantically meaningful correspondences are retained. When the Softmax function is subsequently applied, entries set to $-\infty$ contribute zero probability, effectively removing their influence on the final attention computation. This process sharpens the model's focus on the most discriminative interdomain relationships, while reducing computational redundancy and stabilizing feature alignment across domains.

After sparse selection and normalization, the retained channel-wise attention responses are used to modulate the corresponding value features, yielding the updated cross-domain representations

$$\mathbf{F}'_{\mathrm{sd}} = \mathbf{A}'_{\mathrm{st}} \odot \mathbf{V}_t, \quad \mathbf{F}'_{\mathrm{td}} = \mathbf{A}'_{\mathrm{ts}} \odot \mathbf{V}_s \tag{5}$$

where $\odot$ denotes element-wise multiplication. In this way, the retained attention responses selectively emphasize informative cross-domain feature channels while suppressing less relevant ones.

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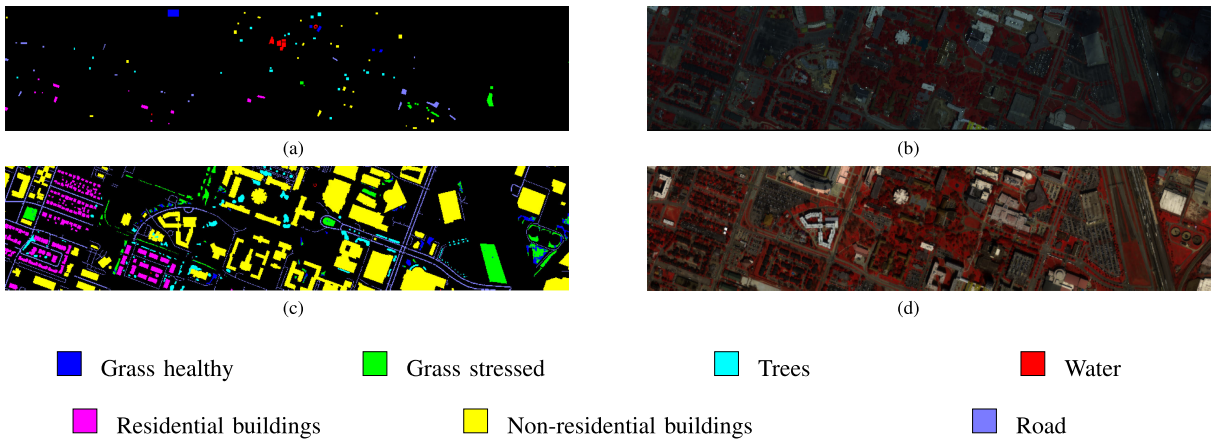


Fig. 4. GT and PCI of Houston task. (a) Ground truth of Houston 2013. (b) Pseudo-color image of Houston 2013. (c) Ground truth of Houston 2018. (d) Pseudo-color image of Houston 2018.

After channel-wise modulation, residual connections and post-attention normalization are applied to obtain the enhanced source and target representations, denoted as $\mathbf{F}_{\text{sd}}$ and $\mathbf{F}_{\text{td}}$, respectively. Through bidirectional cross-domain interaction, each domain representation incorporates complementary information from the other domain, facilitating more effective feature alignment.

Through this design, ISCA enables efficient cross-domain feature exchange while suppressing redundant interactions through sparse attention. As a result, the learned representations become more compact, transferable, and domain-invariant. Furthermore, ISCA complements DSFE by explicitly modeling interdomain relationships, whereas DSFE focuses on enhancing intradomain spectral–spatial representations. Together, they jointly improve feature discrimination and cross-domain alignment for HSIDA.

C. LMMD Gradient-Guided Reweighting

In hyperspectral DA, domain discrepancies often originate from subtle yet uneven variations in spectral characteristics. Such discrepancies may manifest unevenly across the learned feature space, making domain- or class-level alignment alone insufficient. To achieve fine-grained distribution alignment, the proposed LGGR module leverages the LMMD loss both as a domain shift estimator and as a supervisory signal for dimension-wise feature reweighting.

As depicted in the purple dashed LGGR module in Fig. 2, given the source and target feature representations $\mathbf{F}_{\text{sd}}, \mathbf{F}_{\text{td}} \in {}^{B\times D}$, the LMMD loss measures the class-conditional distribution discrepancy between the two domains as

$$\mathcal{L}_{\text{LMMD}} = \sum_{j=1}^{C} \left\| \boldsymbol{\mu}_{\text{s}}^{(j)} - \boldsymbol{\mu}_{\text{t}}^{(j)} \right\|_{\mathcal{H}}^{2} \tag{6}$$

where $\mu_s{}^{(j)}$ and $\mu_t{}^{(j)}$ denote the kernel mean embeddings of class j in the RKHS. Unlike global MMD, LMMD captures localized discrepancy patterns, thereby providing more fine-grained supervision for feature-level analysis.

Building on this formulation, the LMMD loss is explicitly expressed in terms of $\mathbf{F}_{\text{sd}}$ and $\mathbf{F}_{\text{td}}$. Consequently, its gradients with respect to the two feature sets inherently quantify how each feature dimension influences the domain discrepancy. The partial derivatives $\partial\mathcal{L}_{\text{LMMD}}/\partial\mathbf{F}_{\text{sd}}$ and $\partial\mathcal{L}_{\text{LMMD}}/\partial\mathbf{F}_{\text{td}}$ thus serve as dimension-wise sensitivity indicators. A larger gradient magnitude indicates that the corresponding feature dimension has a stronger influence on the LMMD objective and is therefore more sensitive to cross-domain discrepancy. Conversely, smaller gradients suggest that perturbations along that dimension have limited impact on the alignment objective and thus contribute less to domain mismatch.

After obtaining the dimension-wise gradients, LGGR aggregates them at the batch level and fuses the source–target gradient responses into a unified channel-importance vector. Specifically, the gradients from both domains are first averaged along the batch dimension to obtain stable and noise-reduced estimates

$$\begin{aligned} \mathbf{g}_{\mathbf{s}} &= \text{Mean}\left(\left| \frac{\partial \mathcal{L}_{\text{LMMD}}}{\partial \mathbf{F}_{\text{sd}}} \right| \right) \\ \mathbf{g}_{\mathbf{t}} &= \text{Mean}\left(\left| \frac{\partial \mathcal{L}_{\text{LMMD}}}{\partial \mathbf{F}_{\text{td}}} \right| \right). \end{aligned} \tag{7}$$

These two batch-averaged gradients are then combined into a single fused importance vector through a cross-domain aggregation operator

$$\mathbf{g}_{\mathbf{f}} = \frac{\mathbf{g}_{\mathbf{s}} + \mathbf{g}_{\mathbf{t}}}{\text{Mean}\left(\mathbf{g}_{\mathbf{s}} + \mathbf{g}_{\mathbf{t}}\right) + \varepsilon} \tag{8}$$

where the denominator performs global rescaling to prevent dominance by any single dimension and ε is a small constant for numerical stability. The resulting importance vector is detached from the computational graph and used solely for feature reweighting. This unified importance vector is subsequently used to reweight the source and target features in a dimension-wise manner

$$\mathbf{F}_{\text{sd}}^{\text{rw}} = \mathbf{F}_{\text{sd}} \odot \mathbf{g}_{\text{f}}, \quad \mathbf{F}_{\text{td}}^{\text{rw}} = \mathbf{F}_{\text{td}} \odot \mathbf{g}_{\text{f}} \tag{9}$$

emphasizing discrepancy-sensitive feature dimensions while down-weighting dimensions that have limited influence on the alignment objective. Since the gradient magnitude reflects the sensitivity of the LMMD objective to each feature dimension, the resulting reweighting naturally directs the alignment process toward dimensions that are more relevant to cross-domain discrepancy.

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By applying the fused importance vector to both domains, the LGGR module performs dimension-wise reweighting that emphasizes feature channels with stronger LMMD gradient responses while suppressing less informative ones. Since the gradient magnitude reflects the sensitivity of each feature dimension to the LMMD objective, the resulting reweighting naturally guides the alignment process toward discrepancy-sensitive regions of the feature space. Furthermore, the sparse interaction mechanism in ISCA helps suppress redundant feature responses, enhancing the robustness of the reweighted representations. As a result, LGGR enables a more fine-grained and targeted alignment process, yielding more transferable and domain-invariant features. The resulting representations are then fed into both the classification head and the contrastive learning module to support reliable prediction and cross-domain consistency.

D. Progressive Confidence-Weighted Contrastive Learning

To enhance cross-domain discriminability while mitigating the impact of unreliable target-domain pseudo-labels, we introduce a PCCL mechanism. As depicted in the blue dashed PCCL module in Fig. 2, the core idea is to treat prediction confidence as a soft reliability signal and progressively increase its influence over training. This allows early stage optimization to remain stable while gradually incorporating confidence cues as the classifier becomes more reliable.

1) Preprocessing: To enrich instance diversity and increase the number of informative positive and negative pairs, each input sample undergoes semantic-preserving data augmentations. In particular, radiation-noise perturbation and flip augmentation are employed to generate complementary views that preserve class identity while introducing controlled variability. The resulting augmented representation is expressed as

$$\mathbf{F}_{\text{aug}} = \text{Aug}(\mathbf{F}) \tag{10}$$

where Aug(·) represents the augmentation operator.

The original (nonaugmented) source and target samples are further forwarded through the classification head to generate prediction probability vectors. Because the target domain lacks ground-truth supervision, these predicted distributions differ considerably in reliability. Samples located near the decision boundary tend to produce flatter, low-confidence probability vectors, whereas those farther from the boundary yield sharper and more decisive predictions. The maximum predicted probability of each sample thus serves as a natural indicator of model confidence, and the corresponding argmax is adopted as the pseudo-label for target samples. For consistency across domains, the same confidence extraction procedure is also applied to source samples. The resulting source labels and pseudo-target labels are then concatenated to form the unified label vector used to determine positive and negative pairs for contrastive learning

$$\mathbf{Y} = \left[\mathbf{Y}_{\text{s}}, \hat{\mathbf{Y}}_{\text{t}}\right] \tag{11}$$

which is then used to determine positive and negative sample relationships during contrastive learning.

2) Confidence-Weighted Contrastive Loss: Before applying confidence weighting, PCCL first evaluates the semantic similarity between samples by computing pairwise cosine similarity across their feature embeddings, producing a matrix $\mathbf{M}_{\text{sim}}$ that captures the relative proximity of each pair in the latent space. While this measure reflects intrinsic relationships among samples, it is insufficient on its own to guide contrastive optimization reliably under the presence of noisy pseudo-labels in the target domain.

To address this limitation, PCCL introduces a confidence-aware weighting mechanism. The model interprets the maximum predicted probability of each sample as a continuous indicator of its reliability: samples near decision boundaries exhibit smooth, low-contrast probability distributions reflecting high uncertainty, whereas samples far from boundaries produce sharply peaked distributions indicative of confident predictions. Let $\mathbf{P}_{\max}$ denote the vector of maximum probabilities for all samples. PCCL constructs a pairwise confidence matrix via an outer product operation

$$\mathbf{M}_{\text{cf}} = \left(\mathbf{P}_{\max}\mathbf{P}_{\max}^{\top}\right)^{\tau} \tag{12}$$

where each entry $\mathbf{M}_{\text{cf}}(i, j)$ encodes the joint reliability of the sample pair (i, j): large values indicate that both samples are predicted with high confidence, while smaller values signify that at least one sample is uncertain and should exert reduced influence on the contrastive loss. By integrating $\mathbf{M}_{\text{sim}}$ and $\mathbf{M}_{\text{cf}}$, PCCL implements a nuanced, confidence-weighted pairwise optimization that selectively emphasizes semantically relevant and reliable sample pairs, thereby mitigating the adverse effects of pseudo-label noise while preserving informative relationships in the embedding space.

Directly incorporating confidence in early training can, however, be detrimental. As illustrated in the "confidence-guided contrastive learning" module in Fig. 2, a poorly initialized classifier may produce overconfident misclassifications. For instance, the true class corresponds to the second category with a low predicted probability of 0.13, while the model assigns a substantially higher confidence of 0.55 to an incorrect first category. A poorly initialized classifier may thus assign high confidence to incorrect classes, producing spurious high-confidence pairs that severely mislead the optimization. Such erroneous pairs are more harmful than the uniform weighting used in standard contrastive learning, as they may repeatedly reinforce incorrect associations and lock the model into suboptimal decision boundaries. To address this risk, PCCL modulates the influence of confidence through a progressive learning strategy, in which a temperature parameter τ grows linearly with training epoch

$$\tau(e) = \tau_0 + \frac{e}{E}(\tau_1 - \tau_0) \tag{13}$$

where e denotes the current epoch. Notably, the temperature parameter $\tau \in (0, 1)$ governs the concentration of the confidence-induced weighting distribution. When τ approaches 0, the weighting tends toward a uniform distribution, rendering the learning dynamics closer to conventional contrastive learning without confidence modulation. In contrast, as τ approaches 1, the distribution becomes increasingly sharp,

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thereby amplifying the influence of high-confidence samples and steering the model toward a confidence-aware contrastive regime.

From a learning perspective, the evolution of τ can be interpreted as a mechanism that regulates the model's reliance on classifier-derived confidence. In the early training stage, a relatively small τ implies a conservative trust in the classifier, effectively mitigating the adverse impact of over-confident misclassifications, which would otherwise induce spurious high-confidence pairs and destabilize the optimization. Because a smaller τ yields a flatter confidence-weighting distribution, the contribution of any single high-confidence prediction is naturally moderated, reducing the risk that incorrectly confident pseudo-labels dominate the contrastive optimization. As training progresses and the classifier becomes progressively better calibrated, the gradual increase of τ reflects an enhanced trust in its predictions, enabling the model to more fully exploit confidence information and emphasize informative sample pairs. This progressive schedule thus establishes a principled transition from cautious learning to confident discrimination, balancing robustness and adaptivity throughout training.

Finally, PCCL introduces a ConfCL

$$\mathbf{S}(i,j) = \exp\left(\frac{\mathbf{M}_{\text{sim}}(i,j)}{\gamma}\right)$$

$$\mathcal{L}_{\text{ConfCL}} = -\frac{1}{N}\sum_{i=1}^{N}\frac{1}{|P(i)|}\sum_{j\in P(i)}\log\frac{\mathbf{M}_{\text{cf}}(i,j)\,\mathbf{S}(i,j)}{\sum_{k\neq i}\mathbf{M}_{\text{cf}}(i,k)\,\mathbf{S}(i,k)} \tag{14}$$

where P(i) denotes the positive set of sample i and γ is the standard contrastive learning temperature. This formulation adaptively strengthens reliable sample pairs while suppressing the influence of uncertain ones, thereby preventing the network from being steered toward incorrect decision boundaries and yielding more robust cross-domain discriminative representations.

IV. Experiment

In this study, three widely used cross-scene HSI datasets—Houston 2013–2018, Pavia University–City Center (PaviaU–C), and Shanghai–Hangzhou—are employed to comprehensively evaluate the effectiveness and generalization ability of the proposed method. These datasets exhibit substantial variations in spatial resolution, spectral characteristics, land-cover composition, and acquisition conditions, thus providing a challenging benchmark for UDA in HSIC.

A. DA Task Description

1) Houston Task: The Houston datasets consist of two HSIs acquired over the University of Houston campus in different years, namely Houston 2013 [45] and Houston 2018 [46]. The Houston 2013 dataset, which is adopted as the source domain in our experiments, contains 349 × 1905 pixels with a spatial resolution of 2.5 m and 144 spectral bands, among which 48 effective bands are retained after discarding noisy channels. The Houston 2018 dataset serves as the target domain and includes 209 × 955 pixels with a finer spatial resolution of 1 m, providing the same set of 48 spectral bands. Pixel-level annotations for seven common land-cover classes are available for both domains, and representative pseudo-color images together with the corresponding ground-truth maps are illustrated in Fig. 4.

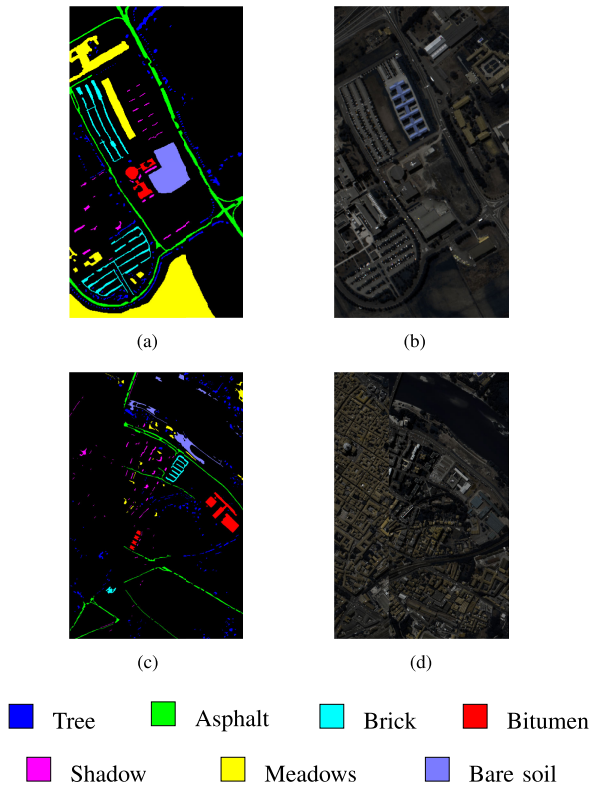


Fig. 5. GT and PCI of Pavia task. (a) Ground truth of PU. (b) Pseudo-color image of PU. (c) Ground truth of PC. (d) Pseudo-color image of PC.

2) Pavia Task: The Pavia datasets are acquired by the ROSIS sensor and include two HSIs, referred to as Pavia University and Pavia Center. The Pavia University comprises 610 × 340 pixels with 103 spectral bands, where the last band is removed following common practice to match the spectral dimensionality of the Pavia Center dataset, resulting in 102 effective bands. The Pavia Center contains 1096 × 715 pixels with 102 spectral bands. Pixel-level annotations for seven shared land-cover categories are provided for both images. The representative pseudo-color images together with the corresponding ground-truth maps are illustrated in Fig. 5.

3) S-H Task: Shanghai and Hangzhou hyperspectral scenes are acquired by the EO-1 Hyperion sensor. After removing water-absorption and noisy channels, both scenes retain 198 spectral bands. The Shanghai scene includes 1600 × 230 pixels, while Hangzhou scene consists of 590 × 230 pixels. Pixel-level annotations for three shared land-cover classes are available for both scenes. The visualizations of the scenes along with their ground-truth annotations are shown in Fig. 6.

B. Experimental Setting

To comprehensively evaluate the effectiveness of PC2L-Net, we compare it with eight representative hyperspectral DA methods covering different adaptation paradigms. DANN [10]

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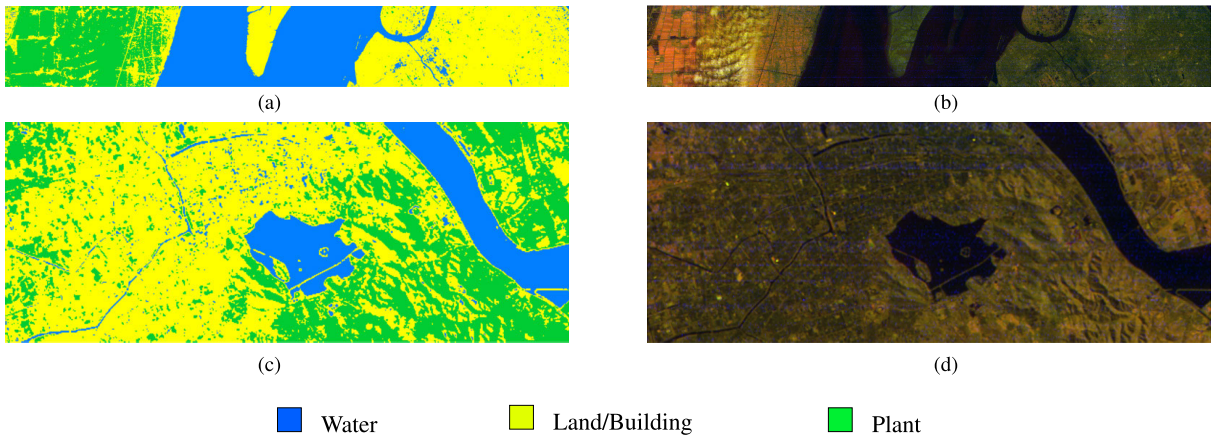


Fig. 6. GT and PCI of S-H task. (a) Ground truth of Shanghai. (b) Pseudo-color image of Shanghai. (c) Ground truth of Hangzhou. (d) Pseudo-color image of Hangzhou.

TABLE I
CLASSIFICATION RESULTS FOR THE HOUSTON TASK

Class	DANN	DSAN	TSTnet	CLDA	MLUDA	CSTnet	S4DL	DAMamba	Ours
1	79.66	**93.71**	85.09	73.73	66.19	81.97	75.62	60.90	77.22
2	74.09	46.55	68.73	**92.43**	76.63	75.51	72.01	75.72	78.82
3	65.00	59.67	54.82	70.68	57.41	59.01	**72.78**	62.39	58.97
4	**100**	59.09	**100**	70.91	89.09	**100**	77.27	54.55	96.82
5	59.12	76.61	66.75	82.64	79.80	77.50	37.44	63.30	**90.47**
6	76.01	72.07	83.01	54.74	81.78	81.64	**82.95**	81.91	78.00
7	54.95	56.23	58.65	**80.83**	52.40	53.23	65.95	61.12	66.56
OA(%)	71.18±1.37	68.19±3.07	75.74±4.40	72.95±2.09	75.91±1.67	76.10±3.05	74.61±2.09	75.41±1.84	**76.93±1.40**
AA(%)	72.69±2.85	66.28±2.93	73.86±1.81	75.14±2.22	71.90±1.74	75.55±2.65	69.14±4.97	63.10±1.94	**78.12±1.34**
Kappa	55.33±4.07	52.31±4.96	60.27±3.57	54.21±2.62	61.30±1.44	61.48±2.33	61.79±2.88	62.81±2.19	**64.33±1.70**

and DSAN [36] represent widely adopted adversarial and discrepancy-based alignment approaches. CLDA [47] introduces confidence-aware pseudo-label learning, while MLUDA [44] performs multilevel alignment across image, feature, and semantic spaces. TSTnet [48] exploits topological graph structures for cross-DA, and CSTnet [49] employs content–style disentanglement to improve domain invariance. In addition, two recent state-of-the-art methods, S4DL [50] and DAMamba [51], are included for comparison. S4DL develops a shift-sensitive spatial–spectral disentangling learning framework for hyperspectral UDA and achieves competitive cross-scene hyperspectral classification performance. DAMamba leverages the strong long-range modeling capability of state-space models to enhance cross-domain hyperspectral representation learning.

Following the standard UDA setting, only the source-domain samples are labeled, while the target-domain samples remain unlabeled during training. Specifically, for each source-domain class, 180 labeled samples are randomly selected for training, whereas all target-domain samples are used as unlabeled data for DA. All models are trained for a fixed number of epochs, and no target-domain labels are used for model selection. The experiments are repeated ten times with different random seeds, and the average results are reported.

All experiments are conducted using the PyTorch framework (Python 3.11.9) on an NVIDIA RTX 4060 GPU. Performance is evaluated using overall accuracy (OA), average accuracy (AA), and the Kappa coefficient.

C. Experimental Results

1) Houston Task: For the Houston task, Table I shows that our framework attains the highest OA, AA, and Kappa among all competing methods. Compared with the recent strong baselines TSTnet and CSTnet, PC2L-Net improves OA by 1.19% and 0.83%, respectively, and raises AA by 4.26% and 2.57%. This substantial gain in class-balanced accuracy is largely attributed to the LGGR module, which adaptively modulates the class-wise contributions during domain alignment, enabling the model to better learn minority or difficult classes. Furthermore, our framework surpasses the traditional contrastive learning-based approach MLUDA by a clear margin across all metrics. This advantage stems from the proposed PCCL, which integrates prediction confidence into the contrastive objective, alleviating the detrimental influence of unreliable pseudo-labels—a limitation that often constrains conventional contrastive learning in UDA.

The classification maps in Fig. 7 further corroborate these observations. As highlighted by the white rectangle, the cross-road region in the Houston 2018 scene presents a particularly challenging case due to the coexistence of multiple land-cover types and complex spatial structures. In this area, several

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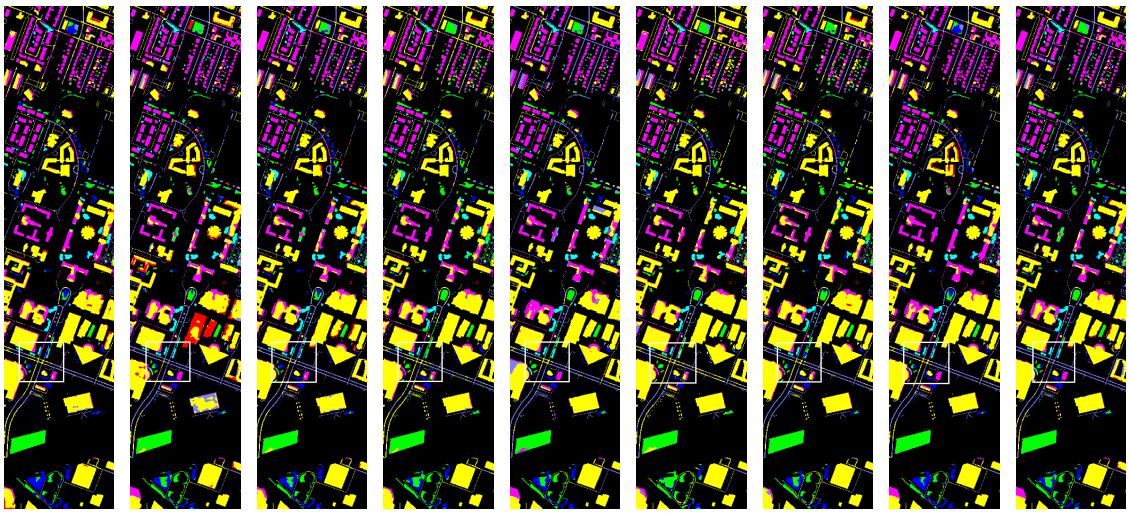


Fig. 7. Classification map of Houston task. (a) DANN, (b) DSAN, (c) TSTnet, (d) CLDA, (e) MLUDA, (f) CSTnet, (g) S4DL, (h) DAMamba, and (i) our framework.

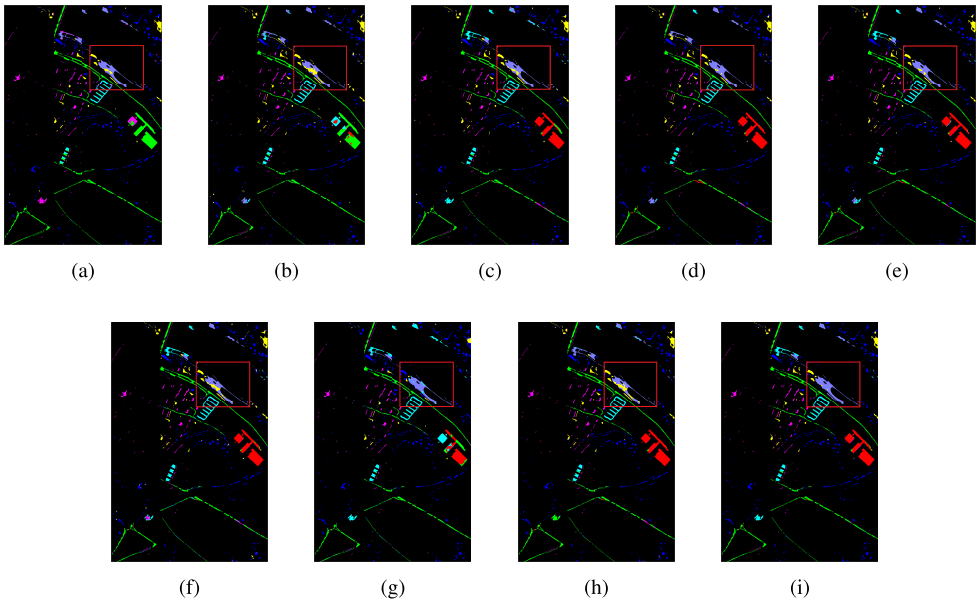


Fig. 8. Classification map of Pavia task. (a) DANN, (b) DSAN, (c) TSTnet, (d) CLDA, (e) MLUDA, (f) CSTnet, (g) S4DL, (h) DAMamba, and (i) our framework.

strong recent baselines (e.g., MLUDA, CSTnet, and S4DL) incorrectly classify road pixels (Class 7) as nonresidential buildings (Class 6). In contrast, PC2L-Net accurately preserves the road structure and produces more coherent predictions. This improvement suggests that PC2L-Net can better handle ambiguous regions with complex spatial structures, resulting in more accurate and spatially consistent classifications.

2) Pavia Task: For the Pavia task, the results in Table II show that PC2L-Net achieves the best OA, AA, and Kappa among all competing methods. In particular, it obtains superior performance on several challenging categories, including Brick and Meadows, where the domain discrepancy between Pavia University and Pavia Center is relatively pronounced. Compared with CLDA, which also incorporates prediction confidence during adaptation, PC2L-Net achieves further improvements across all three evaluation metrics. One possible reason is that CLDA relies on a hard confidence threshold to select pseudo-labeled samples, whereas PC2L-Net adopts a progressive weighting strategy that allows uncertain samples to contribute according to their confidence scores. This design enables the model to utilize target-domain information more flexibly while reducing the influence of unreliable pseudo-labels. The classification maps in Fig. 8 further support this observation. As highlighted by the red rectangle, several competing methods misclassify the marked region, whereas PC2L-Net correctly recognizes it as bare soil (Class 7), resulting in a more accurate local classification pattern.

3) S-H Task: For the S-H task, Fig. 9 and Table III demonstrates that our framework maintains a clear advantage under the large intercity shift between Shanghai and Hangzhou. It achieves the best OA, AA, and Kappa, outperforming the strongest baseline MLUDA by 0.97% in Kappa and 0.79%

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TABLE II
CLASSIFICATION RESULTS FOR THE PAVIA TASK

Class	DANN	DSAN	TSTnet	CLDA	MLUDA	CSTnet	S4DL	DAMamba	Ours
1	**98.22**	94.10	92.05	95.78	94.85	96.06	94.09	93.22	96.79
2	95.40	89.06	93.39	95.02	96.02	97.24	**98.02**	94.09	96.96
3	69.04	64.27	72.45	76.22	86.05	82.14	73.27	80.20	**96.61**
4	76.27	85.93	73.14	85.26	**86.18**	83.07	64.06	77.38	82.75
5	99.81	99.59	99.76	**100**	99.56	99.89	99.36	99.24	99.83
6	24.44	63.37	74.56	90.20	85.69	75.03	74.70	69.75	**94.44**
7	86.67	84.33	**92.04**	86.96	83.08	87.49	84.49	91.39	77.40
OA(%)	83.89±1.85	85.72±0.63	86.71±2.71	90.71±1.51	90.57±0.72	90.18±0.81	85.29±1.54	87.89±2.04	**90.95±0.65**
AA(%)	78.55±3.09	82.95±1.88	85.34±3.71	89.92±2.11	90.20±0.47	88.70±1.01	84.00±0.79	86.47±1.37	**92.11±0.40**
Kappa	82.42±2.79	80.98±1.21	84.87±1.27	86.07±1.45	87.16±0.79	88.52±0.58	87.52±1.01	86.09±1.29	**89.25±0.76**

TABLE III
CLASSIFICATION RESULTS FOR THE S-H TASK

Class	DANN	DSAN	TSTnet	CLDA	MLUDA	CSTnet	S4DL	DAMamba	Ours
1	88.37	92.47	90.26	98.92	99.60	96.23	99.07	97.41	**99.85**
2	86.31	88.61	86.04	87.60	88.25	89.80	88.49	88.09	**90.33**
3	81.24	85.09	94.25	92.47	93.07	84.09	87.12	88.34	91.90
OA(%)	85.14±1.37	88.14±1.63	89.10±0.91	90.62±2.04	91.26±0.77	89.03±1.92	91.72±1.51	91.26±1.90	**92.05±0.86**
AA(%)	85.31±1.75	88.72±2.67	90.18±0.78	93.00±1.52	93.74±1.82	90.04±0.95	91.60±1.19	91.28±1.29	**94.03±0.81**
Kappa	75.57±3.55	79.09±3.31	83.43±1.27	83.26±2.77	85.43±0.95	82.48±1.01	84.97±2.39	83.59±2.12	**86.40±1.43**

in OA. These improvements are notable given that the S-H task contains only three classes, where misclassification of any single region has a more significant effect on the overall performance. PC2L-Net shows particularly strong gains in the "Water" and "Land/Building" class, where spectral diversity within urban structures typically leads to uncertain pseudo-labels. By progressively down-weighting unreliable samples and reinforcing high-confidence semantic relations, PC2L-Net stabilizes the contrastive optimization process and yields more coherent cross-city adaptation. The results confirm that even under severe scene-structure variability, the proposed framework remains highly resilient and maintains strong generalization ability.

D. Ablation Study

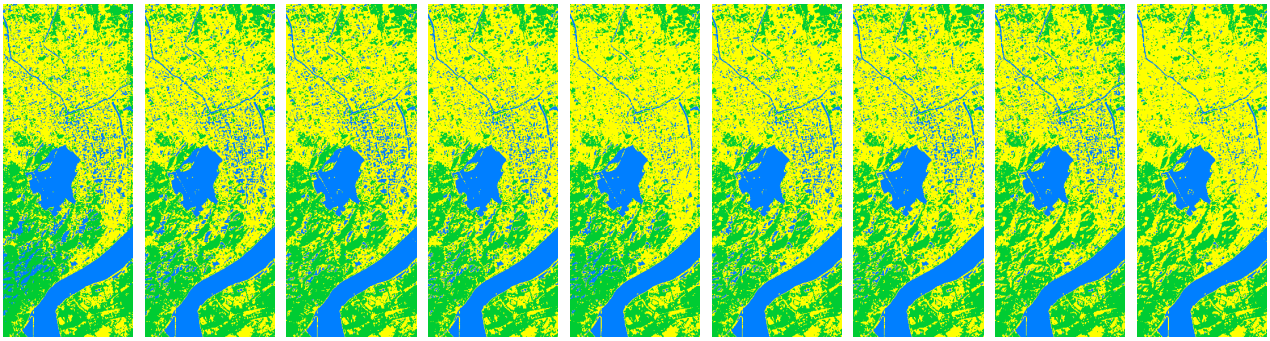
To further validate the effectiveness of each component in our framework, we conduct an ablation study, and the corresponding results are illustrated in Table IV. As shown in the ablation results, the complete model—integrating ISCA, LGGR, and PCCL—achieves the highest OA, AA, and Kappa scores, demonstrating the complementary nature of the three components. When two modules are enabled simultaneously, the model still obtains strong performance, whereas using only a single module leads to relatively limited improvements. Among the individual components, PCCL yields the most notable performance gain, highlighting its critical role in stabilizing and enhancing contrastive learning under UDA, particularly by mitigating the adverse influence of unreliable pseudo-label supervision during training.

TABLE IV
ABLATION STUDY

Task	ISCA	PCCL	LGGR	OA	AA
Houston				69.42 ± 4.44	69.81 ± 5.01
		✓		72.66 ± 2.45	74.15 ± 2.95
	✓	✓		75.01 ± 1.99	75.53 ± 2.18
	✓		✓	73.81 ± 2.87	70.24 ± 3.40
		✓	✓	74.18 ± 1.95	72.11 ± 3.67
	✓	✓	✓	76.93 ± 1.40	78.12 ± 1.34
Pavia				84.71 ± 2.61	86.04 ± 2.04
		✓		86.79 ± 1.99	87.20 ± 1.72
	✓	✓		88.86 ± 1.58	90.42 ± 0.94
	✓		✓	86.94 ± 1.97	87.67 ± 1.75
		✓	✓	87.55 ± 1.41	88.26 ± 1.27
	✓	✓	✓	90.95 ± 0.65	92.11 ± 0.40
S-H				84.93 ± 3.29	84.01 ± 4.04
		✓		87.27 ± 3.01	87.56 ± 3.22
	✓	✓		91.13 ± 2.24	92.09 ± 2.76
	✓		✓	89.03 ± 2.75	89.59 ± 2.16
		✓	✓	89.57 ± 1.77	91.01 ± 2.63
	✓	✓	✓	92.05 ± 0.86	94.03 ± 0.81

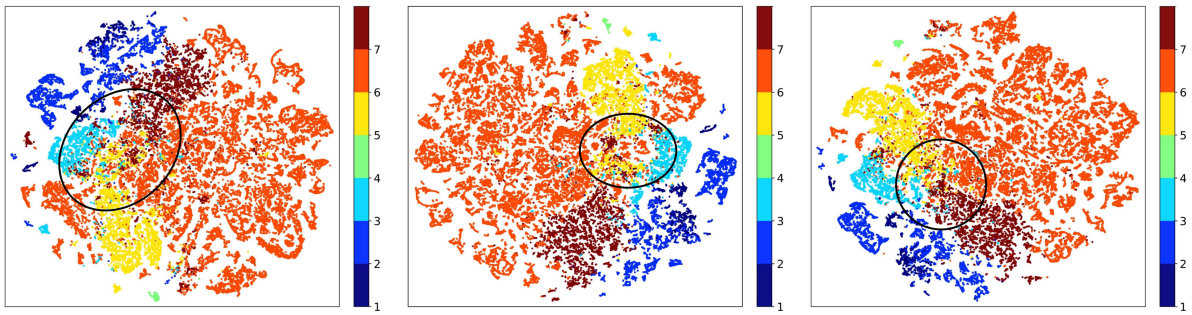
These observations collectively verify that each component provides indispensable benefits. ISCA enhances interdomain feature interaction and strengthens domain-shared representations; LGGR adaptively reweights spectral dimensions based

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(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 9. Classification map of S-H task. (a) DANN, (b) DSAN, (c) TSTnet, (d) CLDA, (e) MLUDA, (f) CSTnet, (g) S4DL, (h) DAMamba, and (i) our framework.



(a) (b) (c)

Fig. 10. Feature visualizations of Houston task. (a) DSAN, (b) MLUDA, and (c) our framework.

on domain discrepancy, facilitating more effective subsequent contrastive learning; and PCCL mitigates the inherent challenges of contrastive learning with noisy pseudo-labels while fully leveraging high-confidence samples to improve cross-domain alignment. Together, these modules work synergistically to produce robust and discriminative representations across domains.

E. Feature Visualizations

Fig. 10 presents the t-SNE visualizations of the learned feature representations for DSAN, MLUDA, and the proposed PC2L-Net, illustrating the progressive effect of different alignment strategies on cross-domain feature organization.

DSAN performs DA without contrastive learning. As highlighted by the black circles in Fig. 10(a), the learned features exhibit relatively dispersed distributions, with limited intraclass compactness and noticeable overlap among different categories. In particular, the fifth category fails to form a coherent cluster, and samples belonging to the same class remain scattered in the embedding space. These observations suggest that distribution alignment alone is insufficient to establish a well-structured discriminative feature space.

By incorporating conventional contrastive learning, MLUDA achieves improved intraclass aggregation. As shown in Fig. 10(b), samples from the same category become more concentrated and the overall cluster structure is more organized than that of DSAN. Nevertheless, the circled region reveals that substantial interclass ambiguity still exists, especially between the sixth and seventh categories, whose feature distributions remain partially intertwined. This indicates that traditional contrastive learning may still be affected by unreliable pseudo-label supervision under domain shift, limiting its ability to achieve fully discriminative representations.

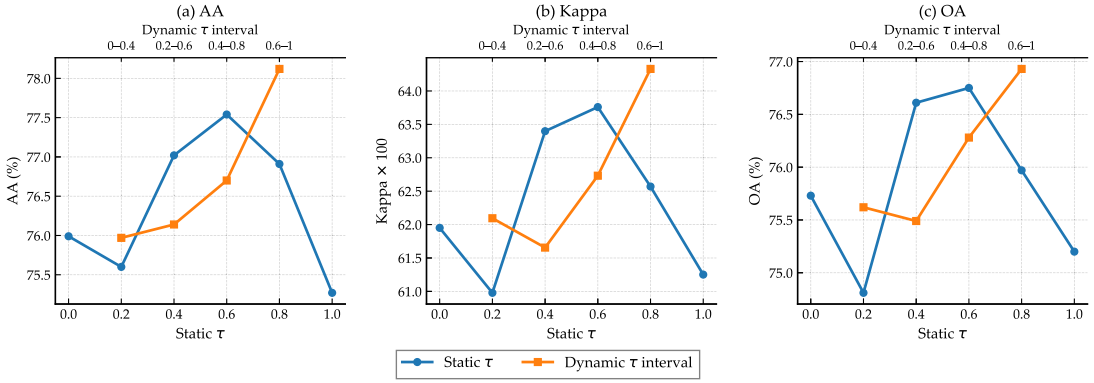
In contrast, PC2L-Net introduces the proposed PCCL mechanism. As highlighted by the black circle in Fig. 10(c), a certain degree of overlap between the sixth and seventh categories still remains. Nevertheless, compared with the heavily mixed distributions observed in MLUDA, the overlap is substantially alleviated, leading to more compact intraclass clusters and more distinguishable interclass boundaries. These results suggest that confidence-guided progressive contrastive learning facilitates the learning of more transferable and discriminative feature representations under domain shift.

F. Parameter Analysis

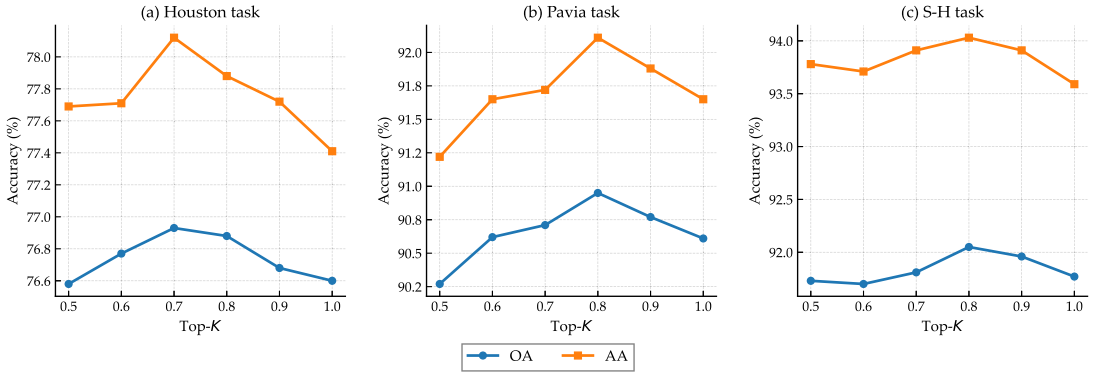
In this section, we investigate the influence of two key hyperparameters in the proposed framework, namely the temperature parameter τ in PCCL and the sparsity ratio ρ in ISCA.

1) Analysis of Temperature Parameter τ: The influence of the temperature parameter τ in PCCL is evaluated on the Houston dataset, and the corresponding results are shown in Fig. 11(a). Without the progressive scheduling strategy, the best performance is achieved when τ is fixed at 0.6. When τ becomes excessively large, the model tends to produce overconfident pseudo-label predictions during training, which

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(a) Temperature parameter τ.



(b) Sparsity ratio ρ.

Fig. 11. Parameter sensitivity analysis of the proposed framework. (a) Influence of the temperature parameter τ in PCCL on the Houston task. (b) Influence of the sparsity ratio ρ in the ISCA module across the Houston, Pavia, and S-H tasks.

may reduce the reliability of confidence-aware weighting. Conversely, a very small temperature weakens the effect of confidence modulation, making the contrastive objective closer to conventional contrastive learning and thereby limiting the contribution of confidence information.

The results further demonstrate the effectiveness of the proposed progressive scheduling strategy. As shown in Fig. 11(a), allowing τ to gradually increase within the interval [0.6, 1.0] consistently yields the best performance. This suggests that a dynamic temperature schedule can better balance the utilization of confidence information at different training stages, leading to more stable contrastive optimization and improved adaptation performance.

2) Analysis of Sparsity Ratio ρ: We further analyze the influence of the sparsity ratio ρ in the proposed ISCA module. Specifically, ρ controls the proportion of channels retained by the Top-K sparse attention operation. The results on the Houston, Pavia, and S-H tasks are presented in Fig. 11(b).

As shown in the figure, the performance generally increases first and then gradually saturates or declines as ρ increases. When ρ is too small, only a limited number of channel interactions are preserved, which may discard useful cross-domain correspondences. Conversely, a large ρ retains more channels but reduces the sparsity of the attention map, making the model more susceptible to redundant or noisy interactions.

For all three tasks, the optimal performance is obtained within a relatively high but still sparse range ($\rho = 0.7$ for Houston and $\rho = 0.8$ for both Pavia and S-H). This observation suggests that preserving a sufficient number of informative channel-wise dependencies is beneficial for cross-domain feature alignment, while excessive sparsification may lead to information loss. Moreover, the performance varies smoothly around the optimal region, indicating that the proposed ISCA module is relatively robust to the choice of ρ across different adaptation scenarios.

G. Complexity Analysis

Table V reports the average execution time per epoch of different methods on the Houston, Pavia, and S-H tasks. Although PC2L-Net introduces additional modules, including PCCL and ISCA, the overall computational overhead remains moderate. Across all three datasets, PC2L-Net requires 4.06, 4.44, and 2.96 s per epoch. The execution time is comparable to recent methods such as MLUDA, S4DL, and DAMamba while remaining lower than that of TSTnet and CSTnet. These results indicate that the performance gains achieved by PC2L-Net do not come at the expense of a substantial increase in training cost. Overall, PC2L-Net achieves competitive computational efficiency while delivering consistent performance improvements across different adaptation tasks.

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TABLE V
EXECUTION TIME (S) FOR ONE EPOCH OF DIFFERENT METHODS

Task	DANN	DSAN	TSTnet	CLDA	MLUDA	CSTnet	S4DL	DAMamba	Ours
Houston	6.56	6.90	8.51	5.12	4.37	8.15	4.94	3.85	4.06
Pavia	7.40	7.01	9.68	5.69	4.49	8.21	5.01	4.31	4.44
S-H	6.62	6.69	6.10	3.50	3.26	7.14	3.37	2.64	2.96

V. CONCLUSION

In this work, we proposed PC2L-Net, a unified and robust DA framework tailored for cross-scene HSIC. The method integrates three complementary components: ISCA, LGGR, and PCCL. Together, these modules improve the discriminative capacity of spectral–spatial representations, emphasize domain-sensitive spectral discrepancies, and effectively mitigate the adverse effects of unreliable pseudo-labels during contrastive alignment. Extensive experiments on three benchmark datasets demonstrate that PC2L-Net consistently exceeds a wide range of classical and state-of-the-art baselines, delivering notable improvements in OA, AA, and Kappa, especially for categories strongly affected by scene shift or ambiguous boundaries. Ablation analyzes further verify the complementary roles of each component, while feature visualizations confirm that PC2L-Net produces tighter intraclass structures and clearer interclass separability, underscoring its effectiveness and robustness for practical HSI DA tasks.

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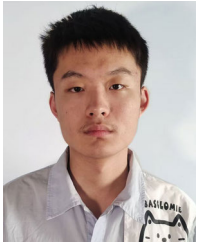
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